A net torque applied to a rigid object always tends to produce:

- ► **rotational equilibrium**
- ► linear acceleration
- ► angular acceleration
- ► rotational inertia

An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:

- ► 12.6 rad/s
- ► 1.57 rad/s
- ► **2.0 rad/s**
- ► 6.3 rad/s

In simple harmonic motion, the restoring force must be proportional to the:

- ► amplitude
- ► frequency
- ► velocity
- ► **displacement**

Mercury is a convenient liquid to use in a barometer because:

- ► it is a metal
- ► it has a high boiling point
- ► it expands little with temperature
- ► **it has a high density**

The units of the electric field are:

- ► **J/m**
- ► J/(C·m)
- ► J/C
- ► J·C

A farad is the same as a

- ► **J/V**
- ► V/J
- ► C/V
- ► V/C

We desire to make an LC circuit that oscillates at 100 Hz using an inductance of 2.5H. We also need a capacitance of:

- ► 1 F
- ► 1mF
- ► 1 µF
- ► **100 µF**

The wavelength of red light is 700 nm. Its frequency is ____________________.

- ► 4.30 * 104 Hertz
- ► 4.30 * 103 Hertz
- ► **4.30 * 105 Hertz**
- ► 4.30 * 102 Hertz

Which of the following statements is NOT TRUE about electromagnetic waves?

- ► Electromagnetic waves satisfy the Maswell's Equation.

- ► Electromagnetic waves can not travel through space.
- ► The receptions of electromagnetic waves require an antenna.
- ► **The electromagnetic radiation from a burning candle is unpolarized.**

Radio waves and light waves are ______________________________.

- ► Longitudinal waves
- ► Transverse waves
- ► **Electromagnetic and transverse both**
- ► Electromagnetic and longitudinal both

Wien's Law states that, *lmax* = ________ K.

- ► 2.90 * 10-3 Hertz
- ► 2.90 * 10-3 s
- ► 2.90 * 10-3 kg
- ► **2.90 * 10-3 m**

Interference of light is evidence that:

- ► the speed of light is very large
- ► light is a transverse wave
- ► **light is a wave phenomenon**
- ► light is electromagnetic in character

Fahrenheit and Kelvin scales agree numerically at a reading of:

- ► **-40**
- ► 0
- ► 273

► 574

According to the theory of relativity:

► **moving clocks run fast**

► energy is not conserved in high speed collisions

► the speed of light must be measured relative to the ether

► none of the above are true

Light from a stationary spaceship is observed, and then the spaceship moves directly away from the observer at high speed while still emitting the light. As a result, the light seen by the observer has:

► higher frequency and a longer wavelength than before

► **lower frequency and a shorter wavelength than before**

► higher frequency and a shorter wavelength than before

► lower frequency and a longer wavelength than before

How fast should you move away from a 6.0 × 1014 Hz light source to observe waves with a frequency of 4.0 × 1014 Hz?

► 20c

► **38c**

► 45c

► 51c

The quantum number n is most closely associated with what property of the electron in a hydrogen atom?

► **Energy**

- Orbital angular momentum
- Spin angular momentum
- Magnetic moment

The quantum number ms is most closely associated with what property of the electron in an atom?

- Magnitude of the orbital angular momentum
- **Energy**
- z component of the spin angular momentum
- z component of the orbital angular momentum

As the wavelength of a wave in a uniform medium increases, its speed will ____.

- Decrease
- Increase
- **Remain the same**
- None of these

The lowest tone produced by a certain organ comes from a 3.0-m pipe with both ends open. If the speed of sound is 340m/s, the frequency of this tone is approximately:

- A. 7Hz
- B. 14 Hz
- C. 28 Hz
- **D. 57 Hz**

To raise the pitch of a certain piano string, the piano tuner:

- **A. loosens the string**
- B. tightens the string
- C. shortens the string
- D. lengthens the string

An object attached to one end of a spring makes 20 vibrations in 10 s. Its angular frequency is:

- **12.6 rad/s**
- 1.57 rad/s
- 2.0 rad/s
- 6.3 rad/s

For an object in equilibrium the net torque acting on it vanishes only if each torque is calculated about:

- the center of mass
- the center of gravity
- the geometrical center
- **the same point**

Ten seconds after an electric fan is turned on, the fan rotates at 300 rev/min. Its average angular acceleration is:

- **3.14 rad/s2**
- 30 rad/s2
- 30 rev/s2
- 50 rev/min2

► 1800 rev/s2

A 4.0-N puck is traveling at 3.0m/s. It strikes a 8.0-N puck, which is stationary. The two pucks stick together. Their common final speed is:

► **1.0m/s**

► 1.5m/s

► 2.0m/s

► 2.3m/s

An object moving in a circle at constant speed:

► must have only one force acting on it

► is not accelerating

► is held to its path by centrifugal force

► **has an acceleration of constant magnitude**

A plane traveling north at 200m/s turns and then travels south at 200m/s. The change in its velocity is:

► 400m/s north

► **400m/s south**

► zero

► 200m/s south

At time t = 0 a car has a velocity of 16 m/s. It slows down with an acceleration given by −0.50t, in m/s2 for t in seconds. It stops at t =

► 64 s

► 32 s

► 16 s

► **8.0 s**

1 mi is equivalent to 1609 m so 55 mph is:

► 15 m/s

► **25 m/s**

► 66 m/s

► 88 m/s

The number of significant figures in 0.00150 is:

► 5

► 4

► 3

► 2

One revolution is the same as:

► 1 rad

► 57 rad

► π/2 rad

► π rad

► 2π rad

For a body to be in equilibrium under the combined action of several forces:

► All the forces must be applied at the same point

► all of the forces form pairs of equal and opposite forces

► any two of these forces must be balanced by a third force

► the sum of the torques about any point must equal zero

A bucket of water is pushed from left to right with increasing speed across a horizontal surface. Consider the pressure at two points at the same level in the water.

► It is the same

► It is higher at the point on the left

► It is higher at the point on the right

► At first it is higher at the point on the left but as the bucket speeds up it is lower there

An organ pipe with both ends open is 0.85m long. Assuming that the speed of sound is 340m/s, the frequency of the third harmonic of this pipe is:

► A. 200 Hz

► B. 300 Hz

► C. 400 Hz

► **D. 600 Hz**

Capacitors C1 and C2 are connected in series. The equivalent capacitance is given by

► C1C2/(C1 + C2)

► (C1 + C2)/C1C2

► 1/(C1 + C2)

► C1/C2

If the potential difference across a resistor is doubled:

- only the current is doubled
- only the current is halved
- only the resistance is doubled
- only the resistance is halved

By using only two resistors, R1 and R2, a student is able to obtain resistances of 3Ω, 4Ω, 12Ω, and 16 Ω. The values of R1 and R2 (in ohms) are:

- 3, 4
- 2, 12
- 3, 16
- 4, 12

Faraday's law states that an induced emf is proportional to:

- the rate of change of the electric field
- the rate of change of the magnetic flux
- the rate of change of the electric flux
- the rate of change of the magnetic field

A generator supplies 100V to the primary coil of a transformer. The primary has 50 turns and the secondary has 500 turns. The secondary voltage is:

- 1000V
- 500V
- 250V
- 100V

Which of the following electromagnetic radiations has photons with the greatest energy?

- ► blue light
- ► yellow light
- ► x rays
- ► radio waves

A virtual image is one:

- ► toward which light rays converge but do not pass through
- ► from which light rays diverge as they pass through
- ► toward which light rays converge and pass through
- ► from which light rays diverge but do not pass through

What is the unit of magnification factor?

- ► meter.Kelvin
- ► radian.Kelvin
- ► degree.Kelvin
- ► no units

During an adiabatic process an object does 100 J of work and its temperature decreases by 5K. During another process it does 25 J of work and its temperature decreases by 5 K. Its heat capacity for the second process is.

- ► 20 J/K
- ► 100 J/K
- ► 15 J/K

- 5 J/K

An ideal gas expands into a vacuum in a rigid vessel. As a result there is:

- a change in entropy
- a decrease of internal energy
- an increase of pressure
- a change in temperature

The Stern-Gerlach experiment makes use of:

- a strong uniform magnetic field
- a strong non-uniform magnetic field
- a strong uniform electric field
- a strong non-uniform electric field

A large collection of nuclei are undergoing alpha decay. The rate of decay at any instant is proportional to:

- the number of undecayed nuclei present at that instant
- the time since the decays started
- the time remaining before all have decayed
- the half-life of the decay

As a 2.0-kg block travels around a 0.50-m radius circle it has an angular speed of 12 rad/s. The circle is parallel to the xy plane and is centered on the z axis, a distance of 0.75m from the origin. The z component of the angular momentum around the origin is:

- **6.0kg · m2/s**

- ► 9.0kg · m2/s
- ► 11 kg · m2/s
- ► 14 kg · m2/s

Questions:

Question No: 1 (Marks: 3)

Two people are carrying a uniform wooden board that is 3.00 m long and weighs 160 N. If one person applies an upward force equal to 60 N at one end, at what point does the other person lift? Begin with a free-body diagram of the board.

Forces in x direction = 0

Forces in Y = F1 + F2 – W

Question No: 2 (Marks: 3)

If a charged particle moves in a straight line through some region of space, can you say that the magnetic field in that region is zero?

Question No: 3 (Marks: 5)

A bike accelerates uniformly from rest to a speed of 7.10 m/s over a distance of 35.4 m. Determine the acceleration of the bike.

Question No: 4 (Marks: 5)

A flat loop of wire consisting of a single turn of cross-sectional area 8.00 cm2 is perpendicular to a magnetic field that increases uniformly in magnitude from 0.500 T to 2.50 T in 1.00 s. What is the resulting induced current if the loop has a resistance of 2.00 W?